

20. API key? What is its purpose? How should it be ~~saved~~ stored and protected, and what could happen if it is exposed publicly?

- An API key is a unique identifier (like a password or token) used to authenticate and authorize access to an API. It tells the API who you are and what you're allowed to do when making requests.

Purpose of an API key

- Authentication
- Authorization
- Usage Tracking

How should it be stored and protected

- Never hard-code in public code repositories
- Use environment variables
- Encrypt keys
- Rotate keys regularly
- Restrict permissions

Risks if exposed publicly

- Unauthorized access
- Data breach
- Financial loss
- Service Abuse.

21. What is a REST ~~API~~ API? What does REST stand for and what are the main HTTP methods (Get, Post, Put, Delete) used in REST APIs? Briefly describe each method.

- A REST API is a style of web service that allows different systems to communicate over the internet using standard HTTP methods. It follows REST principles, which emphasize simplicity, scalability and stateless communication.
- REPRESENTATIONAL STATE TRANSFER - A set of architectural guidelines for designing networked applications

- Main HTTP Methods in REST APIs

Get

- Purpose: Retrieve data from the server

Post

- Purpose: Send new data to the server to create a resource.

Put

- Purpose: Update or replace an existing resource.

Delete:

- Purpose: Remove a resource from the server

2.2 WHAT IS JSON (JavaScript Object Notation)? Why is it commonly used when working with APIs and data exchange? Write a simple example of a JSON object.

- JSON is a lightweight, text-based format used in representing structured data. It stores information as key-value pairs and supports nested structures like arrays and objects. JSON is both human readable and machine-friendly, making it one of the most popular formats for data exchange.

- Commonly used when working with APIs and data exchange?

- Simplicity
- Universality
- Flexibility
- Efficiency
- Integration

23 What is prompt engineering? Why is it an important skill when working with AI tools like ChatGPT / Claude...? 2 examples of a poorly written prompt and an improved version of each

Prompt Engineering is the practice of carefully designing and refining the instructions (prompts) you give to AI systems like ChatGPT or Claude to get the most accurate useful and creative responses. Since these models respond based on how you phrase your request, the quality of your prompt directly affects the quality of output.

Importance.

- ① Precision
- ② Efficiency
- ③ Control
- ④ Professional Use

Example of Poor vs Improved Prompts

- Poor Prompt:

"Explain finance"

- Improved Prompt

Explain the basics of corporate finance to a beginner, using simple examples like household budgeting to illustrate concepts such as capital, debt and investment.

24. What is a MAP Function? Explain what it does in programming and data processing. Provide a simple example in either Python / JavaScript...

- In programming and data processing, a map function is a higher-order function that applies a given operation (function) to each element in a collection (like a list, array, or dataset) and returns a new collection with the transformed results.

How it works:

- You provide a function - the rule transformer
- the map function applies that rule to every item in the list/array
- It returns a new list/array with the updated values.

25. What is Cloud Computing? Name the 3 main cloud service models (IaaS, PaaS, SaaS) and briefly explain each, Give one example of a product for each model.

- Cloud Computing is the delivery of computing services - such as servers, storage, databases, networking, software and analytics - over the internet ("the cloud"). Instead of owning physical hardware or software, organizations and individuals can access these resources on-demand, paying only for what they use.

3 Main Cloud Service Models

IaaS

- definition: Provides virtualized computing resources over the internet (servers, storage, networking)
- Example Product: Amazon Web Services

PaaS

- definition: Provides a platform with tools and frameworks for developers to build, test, and deploy applications without managing infrastructure
- Example: Google App Engine

SaaS

- definition: Delivers software applications over the internet, accessible via a browser or app, without installation or maintenance
- Example: Microsoft 365 (Word, Excel, PowerPoint online) ← cloud-based productivity tools.

26. Version Control System? What is Git and why is it widely used by data analysts and developers? Briefly explain what a commit and a 'branch' are.

VCS (Version Control System) is a tool that helps developers and analysts track changes to files, code, or datasets over time. It allows multiple people to collaborate, manage revisions, and roll back to earlier versions if needed.

Git - the most widely used version control system today. It's popular among developers and data analysts include:

- Tracks every change made to code/data
- It supports collaboration across teams
- It's distributed, meaning every user has a full copy of the project history
- It integrates with platforms like Github, Gitlab and bitbucket for cloud-based collaboration

27. Jupyter Notebook - What is programming language is it most commonly used with? Why is it popular among data analyst and data scientists?

- A jupyter notebook is an open-source interactive environment that allows you to write and run code, visualize data and include explanatory text all in one document. It's widely used in data science, analytics, and machine learning because it combines coding with documentation and visualization seamlessly.

- Programming language
 - Jupyter supports multiple languages but it is mostly common used with Python
 - Popular Among Data Analysts and Data Scientists.
 - Interactive Coding
 - Data Visualizations
 - Documentation
 - Reproducibility
 - Experimentation

28. What is Python? Why is it considered one of the most popular programming languages in Data Analytics and AI? Name 4 Python libraries commonly used in data work and briefly describe each.

- Python is a high-level interpreted programming language known for its simplicity, readability and versatility. It's widely used in data analytics, AI, machine learning, web development and automation because of its rich ecosystem of libraries and ease of learning.

29. Data f
GDPR?
countries

- Data privacy information disclosure
- The GDPR Regulation landmark across all European

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30. DATA E
use of d
each .e

- Data ethics principles
Stored,
Fairness,
respect
about

29. Data Privacy? Why is it important?
GDPR? Explain what it is and which
countries it applies to?

- Data privacy is the protection of personal information from misuse, unauthorized access / disclosure.
- The GDPR (General data Protection Regulation) is the European Unions landmark data privacy law, applicable across all EU member states and the European Economic Area (EEA).

30. DATA ETHICS, 3 Examples of unethical
use of data / AI and explain what makes
each example problematic?

- Data ethics is the framework of moral principles that guide how data is collected, stored, analyzed, and used. It emphasizes fairness, transparency and accountability and respect for human rights. Data ethics is about doing what is right.